<Limits>

$$\lim_{x\to 0^+}(\sin x)^x \Rightarrow (0)^0 \text{ I.F}$$

$$(\sin x)^x = e^{\ln(\sin x)^x} = e^{x\ln\sin(x)}$$

$$L = x\ln(\sin x) = \frac{\ln(\sin x)}{\frac{1}{x}} = \frac{-\infty}{+\infty} \text{ I.F}$$

L'Hôpetal Rule:

$$\frac{\frac{\cos x}{\sin x}}{-\frac{1}{x^2}} = \frac{\cos x}{\sin x}\cdot\frac{-x^2}{1}$$

$$= \frac{-x^2\cdot\cos x}{\sin x} = \frac{(-x\cdot x)\cdot\cos x}{\sin x} = \frac{x}{\sin x}\cdot(-x)(\cos x)$$

We Know: $\frac{x}{\sin x} = 1 \Rightarrow 1\cdot(-x)(\cos x)$

$= 1\cdot(0)(1) = 0$

$$\Rightarrow \lim_{x\to 0^+}(\sin x)^x = e^0 = \boxed{1}$$

E-Vex

$$\lim_{x \to 0^+} (\tan x)^x \rightarrow (0)^0 \quad I.F$$

$$(\tan x)^x = e^{\ln(\tan x)^x} = e^{x \ln(\tan x)}$$

$$L = x \ln(\tan x) = \frac{\ln(\tan x)}{\frac{1}{x}} = \frac{-\infty}{+\infty} \quad I.F$$

$\Rightarrow$ L'Hôpital Rule (L)

$$\frac{\frac{\sec^2 x}{\tan x}}{-\frac{1}{x^2}} = \frac{\frac{\frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x}}}{-\frac{1}{x^2}} = \frac{\frac{1}{\cos x \cdot \sin x}}{-\frac{1}{x^2}}$$

$$\Rightarrow \frac{1}{\cos x \cdot \sin x}\left(-\frac{x^2}{1}\right) = \frac{-x^2}{\cos x \cdot \sin x}$$

$$\Rightarrow \frac{(-x \cdot x)}{\cos x \cdot \sin x} = \frac{-x}{\cos x}\left(\frac{x}{\sin x}\right)$$

We Know: $\frac{x}{\sin x} = 1$

$$= \frac{-x}{\cos x}(1) = 0 \Rightarrow L = 0$$

$$\lim_{x \to 0^+} (\tan x)^x = e^0 = 1$$